Florentin LECOMTE

Future Applied Mathematics and Modeling Engineer

01600 Saint Bernard, France
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Website : https://florentinlecomte.github.io/website/

Education

2025 – 2026 : Master 2 Math in Action (MeA) Dual Degree - Université Claude Bernard Lyon 1

— Specialization in Mathematics, Biology, and Medicine

2022 – 2026 : Engineering Student in Applied Mathematics and Modeling - Polytech Lyon

— **2023 – 2026** : Engineering Program
— **2022 – 2023** : Integrated Preparatory Cycle (PeiP C)

2021 – 2022 : PASS (1st year of Medical School) - Faculty of Medicine Lyon-Est

Professional Experience / Internships

Apr – Sept 2026 : Research Internship - University of St Andrews, Scotland, UK

— Mathematical modeling of cancer invasion and metastatic dissemination.

Sept 2024 – Jan 2025 : Research Internship - Institut Camille Jordan, Villeurbanne, France

— Modeling of Alzheimer's disease using ODEs.
— Theoretical analysis using topological degree, proving the existence of equilibria in a non-autonomous system.
— Development of a recognition algorithm for perinuclear ATM protein crowns in Python.

Summers 2023, 2024, 2025 : Crew Member - McDonald's, Belleville-en-Beaujolais, France

— Teamwork, communication, taking orders in English.

Skills

Programming Languages : C/C++, Python, Matlab, R, LaTeX
Languages : English (TOEIC: 890), Spanish (Beginner)

Academic Projects

Darwinian Evolution : Stochastic simulation by implementing the Gillespie algorithm (C++/Python) and evolutionary branching analysis.

Epidemiology : Modeling the spread of rabies using an SEI model (Matlab).

Optimal Control : Trajectory optimization by implementing Pontryagin's minimum principle to minimize quadratic cost functions (Matlab).

High-Performance Computing : Lid-driven cavity simulation via parallel resolution (PETSc) of the Navier-Stokes equations on an A100 GPU (C++).

Climate Modeling : Modeling of thermohaline circulation using ODEs and numerical simulations (Python).

Neural Networks : Development of a convolutional neural network for image recognition (Python).

Publications

P. Amster, F. Lecomte, F. Olivares, L. Pujo-Menjouet. *Exploring a Non-Autonomous Model Through Topological Degree: A Radiobiology Application.* (In preparation)

J. Restier-Verlet, M. L. Ferlazzo, A. Granzotto, J. Al-Choboq, M. Estavoyer, F. Lecomte, M. Bourguignon, L. Pujo-Menjouet and N. Foray. *Accelerated Aging Effects Observed In Vitro after an Exposure to Gamma-Rays Delivered at Very Low and Continuous Dose-Rate Equivalent to 1-5 Weeks in International Space Station.* **Cells** 2024, 13, 1703. `https://doi.org/10.3390/cells13201703`